Text as Data

W6: Human Coding and Document Classification

Yen-Chieh Liao 廖彥傑

2026-10-16

Fall 2026 (115-1) · National Taiwan University (GIND)
NTLDEV 7191 · Fri. 16:30–18:20 · 國發 204
Last compiled: 2026-10-11 22:06:53 (Asia/Taipei)

Learning objectives

- Write coding rules that another person can apply.
- Build training labels and check coder agreement.
- Explain how Naive Bayes and a linear SVM classify documents.
- Evaluate errors on texts held apart from model development.
- Connect classification performance to a substantive research claim.

W4 and W6: the same research task

Which parliamentary contributions substantively discuss energy policy?

W4: dictionary rules	W6: supervised classification
Researcher selects expressions	Researchers label example documents
Matching rules produce predictions	A model learns associations with labels
Human evidence checks the predictions	Human evidence checks the predictions

Both approaches require a clear definition of **energy-policy attention**.

Reading: Grimmer, Roberts & Stewart (2022), Chapters 16–18.

Text as Data: the W6 reading route

Chapter	Role in today's lecture
17: An Overview of Supervised Classification	Connect labels, classification, and evaluation
18: Coding a Training Set	Develop the codebook and human labels
19: Classifying Documents with Supervised Learning	Explain Naive Bayes and learned classifiers
20: Checking Performance	Design validation and inspect errors

Focus: Chapter 18 for annotation and Chapter 20 for evaluation.

Grimmer, Roberts & Stewart (2022), Text as Data, Part IV.

Empirical cases in Chapters 17–20

Book section	Study and measurement task
17.1: Discursive Governance	Gillion (2016): identify explicit discussion of racial and ethnic minority concerns
18.2.7: Making the News	Boydstun (2013): hand-code newspaper stories by policy topic
19.3: Estimating Jihad Scores	Nielsen (2017): estimate ideological language from clerics' writings
20.3: Validating Jihad Scores	Nielsen (2017): compare scores with independent substantive evidence

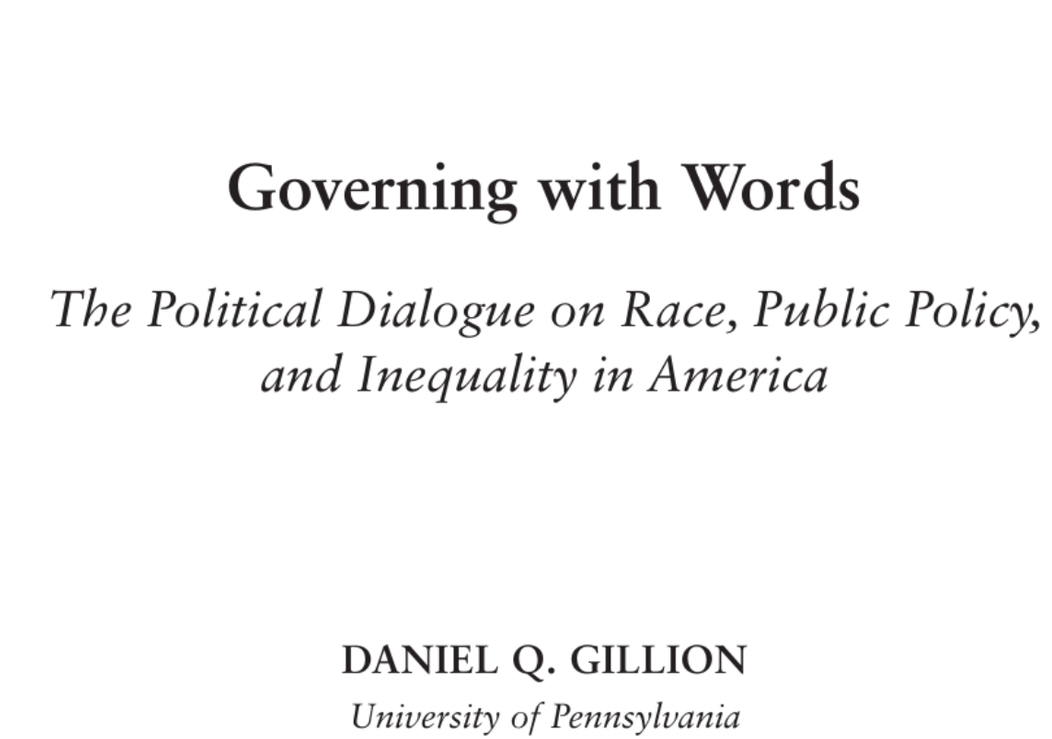
Each case connects a **research question** to a **measurement decision**.

A supervised classification workflow

1. Define the concept, document unit, and target population.
2. Label a sample using a documented codebook.
3. Fit a classifier using the training documents and labels.
4. Choose settings using validation data or cross-validation.
5. Evaluate the frozen workflow on an untouched test sample.
6. Assess whether the errors permit the intended comparison.

Reading: Grimmer, Roberts & Stewart (2022), Chapter 17 and §20.1.

Gillion: measuring political discussion of race



Book title page · Cambridge, 2016

Question: how do politicians discuss racial and ethnic minority concerns?

Texts: presidential public remarks and congressional oral remarks.

Application: humans label paragraphs; classifiers extend the coding to the full corpus.

Research use: compare political attention across politicians and time.

§17.1. Gillion (2016), title-page preview, Appendix A, Appendix B.

Gillion: coding boundaries and agreement

Unit: paragraphs; exclude written bills and procedural material.

Training labels: two assistants code random samples: 300 paragraphs per presidential administration and 1,000 per congressional session.

The rule requires an explicit racial reference: a group, race-specific policy, minority figure, or race-specific issue.

General crime, education, or health discussion qualifies only with such a reference.

Reported intercoder reliability: 97%. Gillion adjudicates disagreements.

Discussion: What would this measure miss about implicit racial appeals?

Gillion (2016), Appendix A, pp. 163–166. The 97% concerns coders, not classifier accuracy.

The codebook defines the outcome

Unit: one complete speech contribution.

Energy = 1: substantive discussion of energy production, supply, regulation, pricing, affordability, or transition.

Other = 0: no such discussion. Exclude personal vigor, institutional power, unrelated bills, and tear gas.

Uncertain: record the ambiguity and discuss it before finalizing labels.

Reading: Chapter 18, §18.2.1. Course application: W6 energy-policy codebook.

Independent coding: five practice texts

1. Electricity prices are making home heating unaffordable.
2. The minister has the power to appoint judges.
3. Coal-mine closures threaten local employment.
4. Police used tear gas during the protest.
5. We must take stronger action on climate change.

Work alone for 2 minutes: label each text and record a reason.

Then compare with a partner. Which rule needs clarification?

Authored teaching sentences. Apply the full-contribution energy-attention codebook.

Evidence and adjudication

Record	Example
Original labels	Coder A: 1, coder B: 0
Evidence	“Coal-mine closures threaten local employment.”
Reason for disagreement	Energy production versus an employment-only interpretation
Decision	Include mine closures under the written rule

Retain both independent labels and the adjudicated label.

Reading: Chapter 18, §18.2. Course example, not an observed coder disagreement.

Coder agreement: a worked example

Two coders independently label the same 20 documents.

Coder A / Coder B	Energy	Other
Energy	8	2
Other	1	9

- Raw agreement: **17 / 20 = 0.85**.
- Agreement expected from the margins: **0.50**.
- Cohen’s kappa: **(0.85 − 0.50) / (1 − 0.50) = 0.70**.

Reading: Chapter 18, §18.2.5. Hypothetical counts. R checkpoint C07.

Reliability and validity

Reliability: can coders apply the same rule consistently?

Validity: does the resulting label capture the intended concept?

Two coders may consistently label every occurrence of `power` as energy policy. They would agree while misclassifying institutional power.

Inspect agreement **and** the substantive reasons for labels.

Reading: Chapter 15, §15.2 and Chapter 18, §18.2.5.

Annotation quality over time

- Pilot the rules on varied development documents.
- Discuss difficult cases and version the codebook.
- Repeat independent checks during longer coding projects.
- Revisit earlier labels when rule changes affect them.

Coder drift: interpretations can change as coders work through the corpus.

Reading: Chapter 18, §18.2.4–18.2.6.

Boydstun: measuring the news agenda



Publisher catalogue · University of Chicago Press, 2013

Question: how is news attention distributed across policy issues over time?

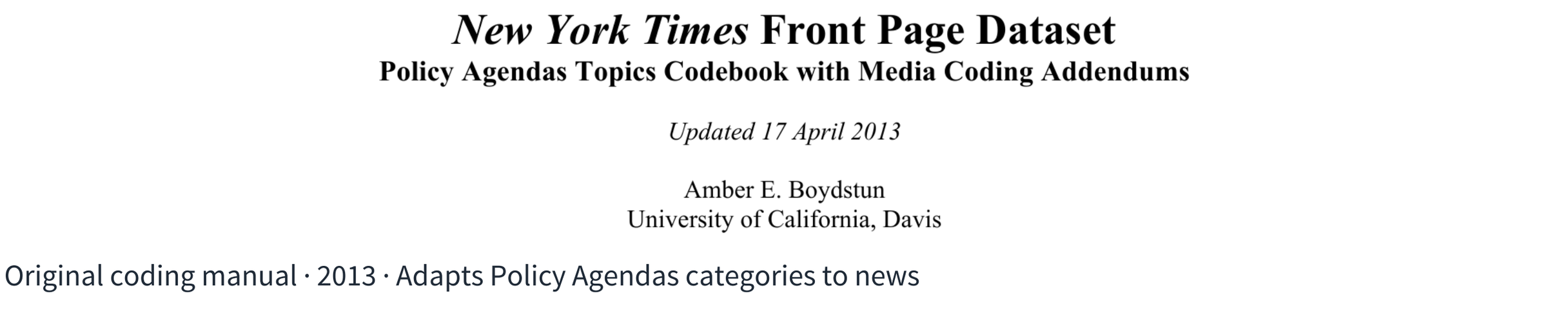
Data: New York Times front-page stories, 1996–2006.

Application: human coders assign policy topics; counts track media attention across issues and time.

W6 connection: construct consistent labels before comparing issues.

§18.2.7. Boydstun (2013), publisher catalogue, printed p. 75, book supplements.

Boydstun: a codebook for newspaper topics



***New York Times* Front Page Dataset**

Policy Agendas Topics Codebook with Media Coding Addendums

Updated 17 April 2013

Amber E. Boydstun
University of California, Davis

Original coding manual · 2013 · Adapts Policy Agendas categories to news

- Read the headline and first three paragraphs.
- Assign one primary topic and record a topic summary.
- Keep articles in the same storyline consistently coded.
- Consult the project director when a decision is unclear.

Lesson: specify the evidence coders may use.

Book connection: §18.2.7. Boydstun (2013), media codebook, pp. 1–5.

Boydstun: labels reveal unequal attention

Table A.1. *Times* front-page attention by topic, 1996–2006, with monthly summary statistics

Topic	Stories	Share of all stories
International affairs and foreign aid	6,354	20.5%
Defense	4,479	14.4%
Health	1,799	5.8%
Energy	299	1.0%

Selected rows; full corpus: **31,034 stories**.

Course diagnostic: an all-“other” energy classifier would have **99.0% accuracy** here, yet detect no energy stories.

Boydstun (2013), Table A.1. Baseline calculated as (31,034 − 299) / 31,034; not a model result in the study.

Sampling a training set

Target: energy attention in parliamentary contributions.

Sampling choice	What it changes
Only texts containing `energy`	Omits other expressions and many negatives
Broad sample across the corpus	Reveals language variation and actual prevalence
Extra rare positive examples	Helps training, but changes the label distribution

Keep an evaluation sample that supports the target population claim.

Reading: Chapter 18, §18.1–18.2.3. Course sampling application.

Training, validation, and test documents

Set	Purpose	Permitted decisions
Training	Estimate model parameters	Learn associations with labels
Validation	Compare candidate workflows	Choose settings and thresholds
Test	Evaluate the frozen workflow	Report performance and remaining errors

Cross-validation can replace a fixed validation set within development data.

Reserve the final test sample before model development.

Reading: Chapter 20, §20.1.1–20.1.2. Cross-validation documentation.

A test of the intended generalization

Intended use	Evaluation design to consider
New contributions from the same population	Random held-out documents
New speakers or institutions	Hold out the relevant groups
Later debates or years	Train earlier and test later

Keep duplicates and closely related text in the same split.

An easy random split may answer a narrower question than the research project.

Reading: Chapter 20, §20.1.4. Grouped and temporal validation.

Cross-validation during development

Example: five folds within the development sample.

1. Fit on four folds and evaluate on the remaining fold.
2. Repeat until each fold has served as validation data.
3. Compare candidate settings using the fold results.
4. Refit the selected workflow on development data.
5. Evaluate once on the separate final test sample.

Fit learned feature selection and weighting inside each training fold.

Reading: Chapter 20, §20.1.2. Cross-validation documentation.

Features and labels

Object	What each row represents
Document-feature matrix, **X**	Word counts for one contribution
Human labels, **y**	The coded category for that contribution
Predictions	Categories inferred for new contributions

Training connects **X** to **y**.

Document IDs keep text, metadata, and labels aligned.

Reading: Chapter 17, §17.2–17.3 and Chapter 19.

Naive Bayes: the main idea

For each category, the model learns how frequently words occur in labelled training texts.

For a new document, it combines:

- A **class prior**: the starting weight for each category.
- **Word evidence**: how compatible its words are with each category.

The multinomial model uses token counts and assumes conditional independence of token draws within a class.

Reading: Chapter 19, §19.1. Naive Bayes.

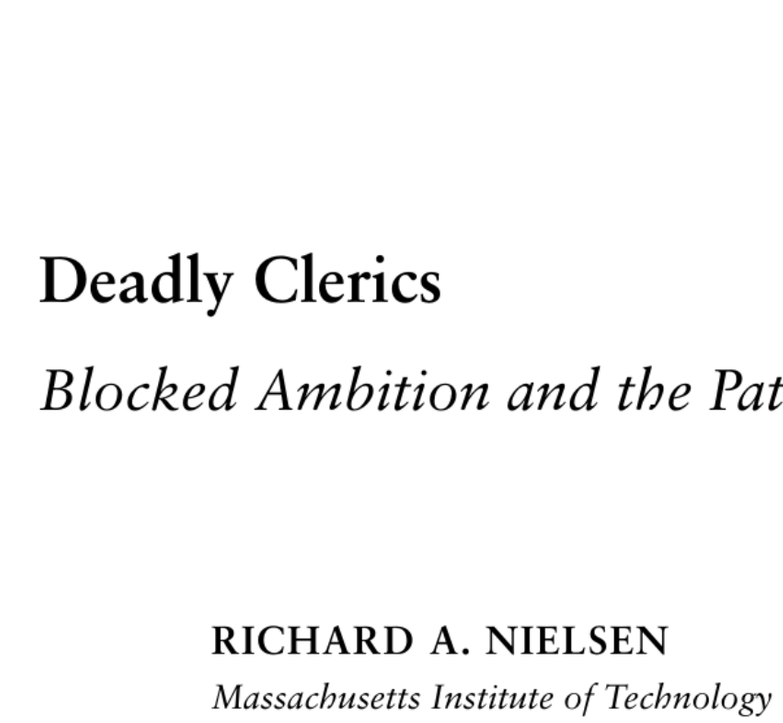
Naive Bayes: assumptions and decisions

Choice	Why it matters
Smoothing	Avoids zero probabilities for known words absent from a class
Class priors	Changes the starting weights of categories
Tokenization	Determines which word forms and phrases are features
Document representation	Determines which information the model receives

Useful classification can coexist with unrealistic language assumptions.

Reading: Chapter 19, §19.1.1. textmodel_nb documentation.

Nielsen: estimating ideological language



Book title page · Cambridge, 2017

Research question: why do some Sunni Muslim clerics adopt militant jihadist ideology?

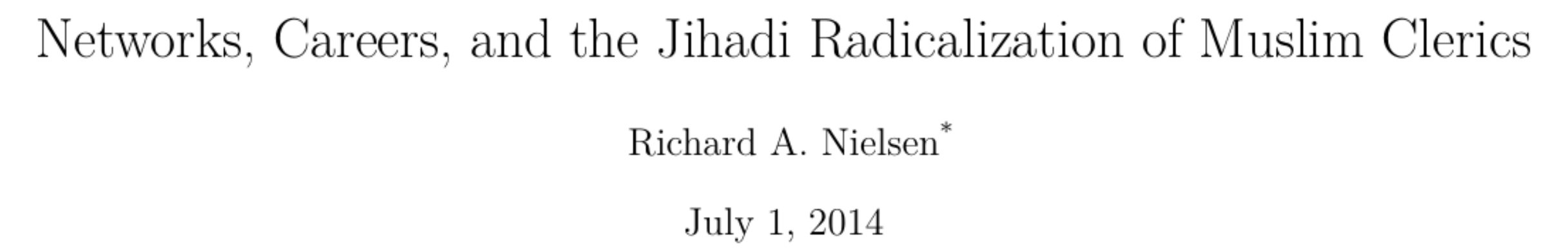
Texts: clerics' Arabic writings, including fatwas, books, and sermons.

Application: turn patterns in language into a cleric-level **jihad score**.

Research use: study ideological differences in relation to networks and careers.

§19.3. Nielsen (2017), Deadly Clerics, title-page preview.

Nielsen: from reference texts to a score



Networks, Careers, and the Jihadi Radicalization of Muslim Clerics

Richard A. Nielsen*

July 1, 2014

Earlier working paper · 2014 · method supplement

- A collection curated by jihadists supplies reference texts.
- A broader sample of clerical writing supplies a comparison.
- Naive Bayes learns contrasting word frequencies.
- The combined writings of each cleric yield a **jihad score**.

The score measures textual resemblance to the reference language. Its ideological interpretation requires validation.

Method supplement: Nielsen (2014), §§4.1–4.2. An earlier working paper, separate from the 2017 book.

Linear SVM: a classification boundary

A linear SVM learns a weighted combination of document features and a boundary between categories.

- The decision score locates a document relative to the boundary.
- Training balances the margin with penalties for classification errors.
- Regularization controls this trade-off.

An SVM decision score requires additional calibration before probability interpretation.

Course extension to Chapter 19, §19.2. Support vector machines.

Classroom R setup

Download W6_classification.R and run one checkpoint at a time.

```
# Install once if needed:
# install.packages(c("quanteda", "quanteda.textmodels", "e1071"))
library(quanteda)
library(quanteda.textmodels)
```

C01: 12 authored training texts, with six labels per category.

C04: six separate teaching test texts with supplied labels.

This small example demonstrates mechanics and errors.

C02–C03: features and Naive Bayes

Run C01 in the R file first to create `train_text` and `train_label`.

```
train_tokens <- tokens(train_text, remove_punct = TRUE)
train_dfm <- dfm(train_tokens)
dim(train_dfm)

nb <- textmodel_nb(train_dfm, y = train_label,
                   smooth = 1, prior = "uniform")
```

Output: 12 documents and 59 features.

The training texts define the feature vocabulary.

Resource: R script. Naive Bayes API.

C04: predicting separate texts

Run C04's data definitions in the R file first.

```
test_dfm <- dfm(tokens(test_text, remove_punct = TRUE))
test_dfm <- dfm_match(test_dfm,
                      features = featnames(train_dfm))
nb_pred <- predict(nb, newdata = test_dfm)
data.frame(id = 1:6, human = test_label, NB = nb_pred)
```

`dfm_match()` aligns the test columns with the training vocabulary.

Words absent from that vocabulary cannot contribute new features.

Resource: R script. dfm_match.

C05: the same features, a linear SVM

```
svm_fit <- e1071::svm(
  x = as.matrix(train_dfm), y = train_label,
  kernel = "linear", cost = 1, scale = FALSE)
svm_pred <- predict(svm_fit,
                    newdata = as.matrix(test_dfm))
```

NB and SVM use the **same labels, vocabulary, and test texts**.

The dense matrix is convenient for this tiny example. Large corpora need an implementation that preserves sparse features.

Resource: R script. e1071 manual.

Reading the actual predictions

Text ID	Human	NB	SVM
1	energy	energy	energy
2	energy	energy	energy
3	other	other	other
4	other	other	other
5	other	other	other
6	**energy**	**other**	**other**

Both models miss the same teaching example.

Executed output from W6_classification.R. Six authored test texts, not a research benchmark.

An error to investigate

“Families cannot afford to keep their homes warm.”

Human: **energy**. NB and SVM: **other**.

Authored test sentence. Executed predictions from the classroom R example.

A confusion matrix

A separate hypothetical test set of 100 documents.

Human / Predicted	Energy	Other
Energy	**18 TP**	**2 FN**
Other	**12 FP**	**68 TN**

- **False positive:** predicts energy for an unrelated contribution.
- **False negative:** misses a contribution about energy policy.

The positive category is **energy**.

Reading: Chapter 20, §20.1. Authored counts for arithmetic, separate from the six-text R demonstration.

Precision, recall, and F1

Metric	Calculation	Interpretation here
Precision	18 / (18 + 12) = **0.60**	60% of predicted energy texts are correct
Recall	18 / (18 + 2) = **0.90**	Finds 90% of human energy texts
F1	2 × 0.60 × 0.90 / (0.60 + 0.90) = **0.72**	Harmonic mean of precision and recall

Accuracy: (18 + 68) / 100 = **0.86**.

Reading: Chapter 20, §20.1. Metric definitions. R checkpoint C06.

Accuracy and rare categories

Suppose only 5 of 100 documents discuss energy policy.

A classifier that predicts **other for every document** has:

- Accuracy: **95 / 100 = 0.95**.
- Recall for energy: **0 / 5 = 0**.

Always compare with a simple baseline and report results for the substantive category.

Authored illustration. Classification metrics.

Classification and estimating a proportion

Return to the hypothetical **100-document** confusion matrix.

Quantity	Calculation
Human energy share	(18 + 2) / 100 = **20%**
Predicted energy share	(18 + 12) / 100 = **30%**
Difference	**+10 percentage points**

The classifier achieves 86% accuracy while overstating energy attention.

Reading: Chapter 19, §19.2.3, “Quantification.” Authored numerical application. R checkpoint C06.

Errors in a comparison between groups

Question: do government and opposition MPs devote different attention to energy?

- Check false positives and false negatives in both groups.
- Inspect changes in vocabulary across periods.
- Report group sample sizes and positive counts.
- Estimate uncertainty before interpreting a difference.

A good pooled score can hide unequal measurement error.

Reading: Chapter 19, §19.2.3 and Chapter 20. Course research-design application.

Validation without a complete human benchmark

Chapter 20 also discusses evidence from:

- Surrogate labels or partial category replication.
- Human review of selected outputs.
- Correspondence with external information.

Each checks part of the interpretation. A surrogate label may capture a different concept.

Document what the available evidence can establish.

Reading: Chapter 20, §20.2.

Nielsen: independent checks on the scores

Combining Interpretivist and Positivist Approaches in Social Science Research*

Jasmine English† and Richard A. Nielsen‡

August 22, 2024

Later methodological discussion · 2024 manuscript

Does the score recover substantive judgments outside its training contrast?

- **Biographies:** do identified jihadist clerics receive higher scores?
- **Experts:** do the Militant Ideology Atlas and Brachman (2009) classifications correspond to the scores?
- **A jihadist image condemning preachers:** are its targets toward the non-jihadist end?

Reported correspondence supports **meaning**, beyond model fit.

Book connection: §20.3. Nielsen (2017), Ch. 5. Nielsen (2014), pp. 15–17; English & Nielsen (2024), pp. 23–24.

Nielsen: the limits of external validation

A missing biographical label is not a confirmed negative.

Some biographies identify jihadist commitments explicitly. Others omit ideology. Silence cannot establish non-jihadist status.

Course discussion: could a text criticizing jihadism receive a high score because it discusses the same vocabulary?

Check passages in context and inspect differences across genres and authors. Independent benchmarks and close reading answer complementary questions.

For our energy task, investigate a discussion of unaffordable heating even when it omits “energy.”

Nielsen (2013), dissertation, §4.4.1, pp. 121–122; Nielsen (2014), p. 17 n. 16. Discussion and energy application are course extensions.

LLM labels and the human benchmark

An LLM can propose labels using the codebook.

- Preserve the prompt, model version, and annotation instructions.
- Compare predictions with independent human judgments.
- Inspect uncertainty, systematic errors, and changes over time.

High agreement between automated methods can reflect shared mistakes.

W12 develops these annotation workflows in more detail.

Course extension. Connects W6 validation principles to the scheduled W12 topic.

Choosing a method for this task

Method	Evidence to examine
Dictionary	Coverage and ambiguity of word rules
Naive Bayes	Quality of labels and fit of count-based features
Linear SVM	Label coverage and the learned decision boundary
LLM annotation	Coding-rule compliance and reproducibility

Compare methods on the **same untouched evaluation sample**.

The task, available labels, and error costs guide the choice.

Reading: Chapter 17 and Chapter 20. LLM row is a course extension.

A methods section for classification

- Concept, document unit, corpus boundaries, and intended use.
- Sampling design, codebook version, coders, and adjudication.
- Original coder agreement and treatment of uncertainty.
- Feature construction, model settings, and split design.
- Test counts, category-specific errors, and uncertainty.
- Remaining limits for the substantive comparison.

Reading: Chapter 15, §15.2 and Chapters 18–20. Course reporting checklist.

Applying the workflow to your project

Discuss in pairs for 3 minutes.

1. What category would you assign to one document?
2. What difficult case needs an explicit coding rule?
3. Which documents would you label for training?
4. What population should the evaluation represent?
5. Which mistake would most distort your substantive claim?

Bring one revised coding rule to the discussion.

Practice and the Week 7 deadline

Practice: run C01–C05 and explain the missed sixth sentence. Use C06–C07 to check the arithmetic by hand.

Optional real-text follow-through: use the existing W4 coding templates and W6 codebook. Keep reserved test documents apart from development.

Week 7, October 23: research title and abstract due.

Describe your concept, text source, and intended validation evidence.

Resources: W6 R script · W4 lab.

Recommended reading priorities

Priority	Sections	Reading question
Core	**18.1–18.2**	What makes training labels usable?
Core	**20.1**	How should we evaluate a classifier?
Overview	17.2–17.5	How do the research steps connect?
Model detail	19.1–19.2	What do NB and learned classifiers assume?
Extension	19.2.3 and 20.2	How do aggregate measures and indirect validation differ?

Chapter 15 supplies additional measurement context.

Grimmer, Roberts & Stewart (2022), Text as Data.

Empirical study references

- **Gillion, Daniel Q. 2016.** *Governing with Words: The Political Dialogue on Race, Public Policy, and Inequality in America*. Cambridge University Press. Appendix A.
- **Boydstun, Amber E. 2013.** *Making the News: Politics, the Media, and Agenda Setting*. University of Chicago Press. Codebook, data, and tables.
- **Nielsen, Richard A. 2017.** *Deadly Clerics: Blocked Ambition and the Paths to Jihad*. Cambridge University Press. Author's book page.

Methods supplements: Nielsen (2014); English & Nielsen (2024).

References and software

Grimmer, J., M. E. Roberts, and B. M. Stewart (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. Chapters 17–20, with Chapter 15 as background. Book and contents.

Software: quanteda for text features, `quanteda.textmodels::textmodel_nb()` for Naive Bayes, and e1071 for the classroom linear SVM.

Evaluation references: scikit-learn documentation on cross-validation and classification metrics.

Energy practice sentences and hypothetical calculation exercises are course materials. Named study results retain their original citations.